

Appropriate Measures

Lesson 8-4

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

Mean of 10, 20, 30 = $(10+20+30) \div 3 = 20$

Mean

The average. Add all the numbers, then divide by how many there are.

Data: 5, 8, 12, 15, 20 → median is 12 (the 3rd of 5 values)

Median

The middle number when you put them in order.

Data: 12, 14, 13, 15, 45 → 45 is far from the cluster, so it is an outlier

Outlier

A number that is much bigger or smaller than the rest.

Scores: 5, 6, 7, 8, 8, 35 → most scores are low, but 35 creates a tail to the right (skewed right)

Skewed

When most data sits on one side with a tail on the other.

Symmetric = even on both sides. Skewed = bunched on one side with a tail

Data distribution

How the data looks: where it sits and how spread out it is.

88, 90, 89, 91 (low variability) vs. 50, 70, 95, 100 (high variability)

Variability

How spread out the numbers are.

Key Ideas & Notes

- The league is creating awards for the season.
- For the scoring title, they need to pick the best measure of a typical game.
- Here are the top scorer's points per game: 22, 24, 20, 25, 23, 21, 58.
- That 58-point game was a record-breaker!

Think About It

- How does the 58-point game compare to the other scores?
- What is the mean? What is the median?
- Which measure is closer to what this player usually scores?

My Notes

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

A top scorer's points were 22, 24, 20, 25, 23, 21, 58. The mean is 27.6 but the median is 23. Which should the league report as the 'typical' game, and why?

Level 1 support

Start here: An outlier can pull the mean away from what is typical, making the median a better choice.

The league should report the ____ because ____.

La liga debería reportar la ____ porque ____.

The 58-point game is an ____ that affects the mean.

El juego de 58 puntos es un ____ que afecta la media.

Word bank: mean median outlier skewed measure of center

Listen for: Students choose the median (23) and identify 58 as an outlier that pulls the mean up to 27.6, making it unrepresentative.

Level 2

Justify: by how much does removing the 58-point game change the mean? What does that show about outliers?

- Without the 58, the mean drops to about ____ because ____.
- This shows the median is ____ to outliers while the mean is ____.

EXPLORE

As you sort scenarios into 'Use Mean' or 'Use Median,' what clue tells you a data set needs the median instead of the mean?

Level 1 support

Start here: Data with an outlier or a skewed shape is better summarized by the median.

I use the median when the data has ____.

Uso la mediana cuando los datos tienen ____.

I use the mean when the data is ____.

Uso la media cuando los datos son ____.

Word bank: **mean** **median** **outlier** **skewed** **measure of center**

Listen for: A strong answer says the presence of an outlier or skew signals the median, while symmetric data with no outliers fits the mean.

Level 2

Compare: give one sports example where the mean is the better choice and one where the median is. Justify each.

- The mean is better for ____ because ____.
- The median is better for ____ because ____.

CONNECT

A reporter writes 'the average playoff ticket is \$85,' but prices are \$45, \$50, \$48, \$52, \$55, \$260. Is \$85 a fair 'typical' price? What measure should the reporter use?

Level 1 support

Start here: When one extreme value skews the data, the median represents the typical case more fairly.

\$85 is / is not fair because the \$260 ticket ____.

\$85 es / no es justo porque el boleto de \$260 ____.

The reporter should use the ____ because ____.

El reportero debería usar la ____ porque ____.

Word bank: mean median outlier skewed measure of center

Listen for: Students explain \$85 is misleading because the \$260 courtside seat skews the mean upward, and the median (~\$51) better reflects a typical ticket.

Level 2

Critique: could the reporter be using the mean ON PURPOSE to make tickets sound pricier? Argue whether that is honest.

- Using the mean here could be misleading because ____.
- An honest report would show ____ so readers understand ____.

REFLECT

For the data set 15, 18, 16, 17, 15, 72, which measure of center best represents the data, and why?

Level 1 support

Start here: An extreme value makes the median the more representative measure of center.

The ____ best represents the data because ____.

La ____ representa mejor los datos porque ____.

The value ____ is an outlier that distorts the mean.

El valor ____ es un valor atípico que distorsiona la media.

Word bank: (mean) (median) (outlier) (skewed) (measure of center)

Listen for: Students choose the median (~16.5), identify 72 as the outlier, and explain it would inflate the mean.

Level 2

Generalize: state a rule for deciding between the mean and the median for any data set.

- Choose the median when the data has ____.
- Choose the mean when ____, because ____.

Guided Examples

Example 1

A gymnast's scores are: 8.5, 8.8, 8.7, 8.6, 8.9. There are no outliers. Which measure best represents a typical score?

Solution: The data is symmetric with no outliers, so the mean (8.7) best represents the typical score.

Answer: A. Mean

Example 2

A runner's mile times are: 7:10, 7:15, 7:12, 7:20, 12:00. The 12:00 was due to a cramp. Which measure best represents a typical mile?

Solution: The 12:00 is an outlier that pulls the mean up. The median (7:15) better represents the runner's typical time.

Answer: A. Median — the outlier 12:00 pulls the mean too high

Example 3

Data set: 5, 6, 7, 7, 8, 50. The mean is about 13.8 and the median is 7. Which better represents the typical value?

Solution: The mean (13.8) is higher than 5 of the 6 values because the outlier 50 pulls it up. The median (7) better represents a typical value.

Answer: A. Median (7) — the outlier 50 inflates the mean

Write About the Math

The Writing Revolution

I can explain my choice using the words mean, median, outlier, and skewed.

1. Kernel Sentence subject + verb

Model: Mean is the average. Add all the numbers, then divide by how many there are.

Media es el promedio. Suma todos los números y divide entre cuántos hay.

Write a kernel sentence about mean. Use a subject and a verb.

Escribe una oración base sobre media. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Mean matters in math

Media importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Mean matters in math because ____.

Media importa en matemáticas porque ____.

but
pero

Mean matters in math, but ____.

Media importa en matemáticas, pero ____.

so
entonces

Mean matters in math, so ____.

Media importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about mean.
Di un hecho verdadero sobre mean.

Mean ____.

Question *Pregunta*

Ask a question about mean.
Haz una pregunta sobre mean.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about mean.
Muestra entusiasmo sobre mean.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with mean.
Dile a un compañero qué hacer con mean.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

The league should report the ____ **because** ____.

La liga debería reportar la ____ *porque* ____.

The 58-point game is an ____ **that affects the mean.**

El juego de 58 puntos es un ____ *que afecta la media.*

I use the median when the data has ____.

Uso la mediana cuando los datos tienen ____.

Try It

Solve on your own. Check the answer key when you are done.

1. A baseball player's batting averages over 6 seasons are: .280, .295, .290, .285, .300, .110. The .110 was an injury-shortened season. Which measure better represents the player's typical batting average?

- A. Median, because the .110 outlier pulls the mean down
- B. Mean, because it uses all the data
- C. Mean, because .110 is a real season
- D. Neither measure works

Show your work:

2. Match the reason with whether to use mean or median.

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A local newspaper reports that the average home price in a neighborhood is \$450,000. The actual prices are: \$180K, \$200K, \$190K, \$210K, \$195K, \$1,725K. Is the reporter's claim misleading? Calculate the mean and median and explain which better represents a 'typical' home price.

Sentence starter: Mean = _____. Median = _____. The reporter's claim is _____ because the outlier (\$1,725K) pulls the mean _____. The _____ (about \$_____) better represents a typical home price because _____.

Show your work:

Reflect — Exit Ticket

Data set: 15, 18, 16, 17, 15, 72. Which measure of center best represents the data?

- A. Median, because 72 is an outlier
- B. Mean, because it uses all values
- C. Mean, because it is always best
- D. Neither works for this data

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. Median, because the .110 outlier pulls the mean down — *The .110 is an outlier that pulls the mean down to .260. The median (.2875) better represents typical performance because it isn't affected by the extreme value.*
2. **Try It 2:**
3. **Exit Ticket:** A. Median, because 72 is an outlier — *The value 72 is an outlier. The mean (25.5) is pulled high by 72 and doesn't represent the typical values. The median (16.5) is a better measure of center.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Mean is the average. Add all the numbers, then divide by how many there are.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).